Toward Reproducible Facial Kinship Verification: A Unified Benchmark with Public and Private Family Datasets

Laith Alkhalil, Mahmoud Al-Ayyoub

***Abstract*—Facial kinship verification remains difficult because related people can differ sharply in age, gender, expression, and capture conditions while sharing only partial inherited traits. Although the literature now spans handcrafted descriptors, transfer learning, metric learning, gated relation models, tri-subject reasoning, and deep verification networks, direct reproduction and comparison are still hindered by fragmented implementations and dataset-specific code. We address that problem by unifying four representative kinship-verification families in one maintained Python toolkit: a classical patch-descriptor baseline, a KinVer-style metric-learning pipeline, native gated autoencoder feature mappers, and a Siamese-style deep model. The framework provides a shared command-line interface, normalized dataset handling, configuration-driven experiments, structured outputs, and regression-tested native implementations. We evaluate the toolkit on KinFaceW-II father-son verification and on a local Families in the Wild (FIW) FIDs export, and we also support a substantial non-public in-house dataset assembled by the first author. On KinFaceW-II, the reproduced KinVer path reaches 85.0% mean accuracy, outperforming the classical baseline at 50.2% and a one-epoch small Siamese model at 49.6% under CPU-only settings. On the local FIW export, the same Siamese model attains 49.75% accuracy but 61.72% area under the precision-recall curve, indicating useful ranking despite weak thresholded decisions. The native gated autoencoder ports are additionally validated on a real KinFaceW matrix, yielding reconstruction errors of 0.6666 and 0.6833 for the standard and multiview variants. The main outcome of this work is a durable experimental benchmark that turns previously disconnected kinship methods into a single extensible research platform.**

***Index Terms*—Kinship verification, facial biometrics, KinFaceW, Families in the Wild, gated autoencoder, metric learning, deep learning, reproducible research.**

I. Introduction

Facial kinship verification asks whether two people depicted in facial images are biologically related. The task is difficult because it lies between identity verification and generic facial similarity. Related people may share family traits without exhibiting the near-identity consistency expected in conventional face recognition, while unrelated people may appear similar because of ethnicity, age, pose, grooming, or capture conditions. The resulting appearance gap makes kinship verification intrinsically challenging, especially under unconstrained settings.

The problem is also practically important. Kinship cues can support family-photo organization, genealogy analysis, social-media mining, missing-child investigation, and broader family-aware search tasks [1]–[4]. More generally, kinship verification is a useful test bed for structured relational reasoning in computer vision because it asks not only who a person is, but whether one face carries inherited evidence about another.

Research on facial kinship has progressed through several methodological waves. Early studies emphasized manually designed features and basic classifiers [1]. Later work introduced transfer learning [5], verification under uncontrolled conditions [3], heredity-aware sparse reconstruction and family classification [6], gated autoencoder relation models [7], metric learning and multiview fusion [8], [9], tri-subject verification [10], and deep large-scale benchmarks such as FIW [4], [11], [12]. Surveys now describe kinship verification as a sufficiently mature subfield to warrant comparative analysis and synthesis [13], [14].

Despite this progress, reproducibility remains a major obstacle. Many kinship implementations survive only as disconnected legacy codebases, research-prototype Python scripts, or benchmark-specific projects with fragile runtime assumptions. This makes it difficult to determine which methods still run, which datasets they expect, and whether the resulting metrics are directly comparable.

This paper addresses that gap by unifying representative kinship-verification families inside one maintained Python toolkit. Instead of proposing yet another isolated algorithm, we package classical descriptors, KinVer-style metric learning, native gated autoencoders, and a Siamese deep runner behind a consistent command-line, configuration, data-loading, and reporting surface. In addition, the toolkit now supports a substantial non-public in-house dataset assembled from scratch by the first author. Although this collection is not yet part of a formal public benchmark protocol, it broadens the scope of the work through age-centric subsets, named family archives, and identical-twin material.

The main contributions of this paper are as follows:

- We migrate four representative kinship-verification families into a unified Python codebase: classical handcrafted baselines, KinVer-style metric learning, native gated autoencoders, and a Siamese deep model.
- We standardize experiment execution, dataset resolution, artifact logging, and reporting so that different kinship methods can be reproduced through a single interface.
- We validate the maintained toolkit on KinFaceW-II father-son verification, on a local FIW FIDs export with real metadata-to-image inconsistencies, and on native GAE matrix inputs.

- We document support for a private in-house kinship collection containing 10 subsets, 220 inferred family groups, 805 people, and 2,318 images, including age-variant families and identical twins.

II. Related Work

A. Early Computational Kinship Verification

Fang et al. first framed kinship verification as a machine-learning problem over parent-child facial image pairs, using handcrafted facial cues and a controlled dataset of 150 pairs [1]. Xia et al. later introduced transfer learning to bridge the appearance gap between parents and children [5], while Zhou et al. studied kinship verification under uncontrolled imaging conditions [3]. These works established kinship verification as a distinct visual problem rather than a simple variant of identity matching.

Kinship reasoning also expanded beyond pairwise verification. Xia et al. studied kin relationships in photographs as a broader relational task [2], and Fang et al. introduced Family101 and heredity-aware sparse reconstruction for kinship classification across families [6]. These studies emphasized that kinship cues are localized, compositional, and family-structured.

B. Relation Learning, Metric Learning, and Deep Models

Dehghan et al. proposed a gated-autoencoder framework to determine which parent an offspring resembles more [7]. Their work is especially relevant here because it motivates the native GAE family preserved in our toolkit. Lu et al. proposed neighborhood repulsed metric learning (NRML) and multiview NRML for kinship verification [8], while Hu et al. explored large-margin multi-metric learning for face and kinship verification in the wild [9]. These methods demonstrated that kinship benefits from task-specific learned projections and complementary feature fusion.

Qin et al. extended the problem to tri-subject verification, explicitly modeling the child-parents unit rather than only a pair [10]. More recently, large-scale deep kinship research has been driven by FIW and related benchmarks [4], [11], [12], as well as deep metric-learning approaches such as KinNet [15]. Reviews by Qin et al. and Wu et al. provide a broader synthesis of the field and its open challenges [13], [14].

C. Benchmark Datasets

Dataset evolution has strongly shaped kinship-verification research. Early datasets were relatively small and relation-specific [1], [16]. KinFaceW-I and KinFaceW-II became standard public pair-verification benchmarks under web conditions [8]. Family101 broadened the scope to family classification [6]. TSKinFace introduced tri-subject evaluation [10]. FIW later became the largest and most diverse public kinship benchmark, organizing family trees and multiple kin relations at scale [4], [11].

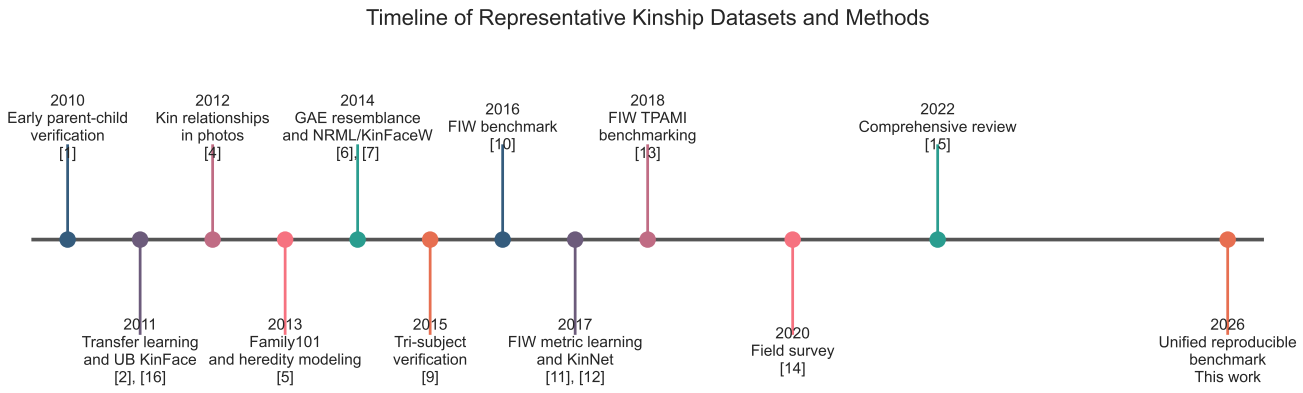

Fig. 1. Representative timeline of kinship datasets and methods.

TABLE I
BENCHMARK DATASETS RELEVANT TO THE PRESENT WORK AND TOOLKIT CONTEXT.

Dataset	**Citation**	**Main Structure**	**Scale / Notes**	**Relevance to This Work**
Cornell parent-child dataset	[1]	Pair verification	150 parent-child pairs collected from public figures	Historical handcrafted baseline reference
UB KinFace	[16]	Pair verification	Early genealogical kinship dataset	Important pre-KinFaceW benchmark
KinFaceW-I / KinFaceW-II	[8]	Pair verification	Four parent-child relations under web conditions	Primary public benchmark used in this paper
Family101	[6]	Family classification	101 public families, 14,816 images	Related benchmark and heredity modeling reference
TSKinFace	[10]	Tri-subject verification	Child with both parents	Important family-structured benchmark
FIW	[4], [11]	Family tree benchmark	Large-scale, multi-relation, in-the-wild kinship database	Large-scale benchmark used here via local FIDs export
Private in-house dataset	This work	Family, age-centric, and twin subsets	10 subsets, 220 inferred families, 805 people, 2,318 images	Non-public research asset supported by the toolkit

III. Unified Toolkit and Research Assets

The unified toolkit was designed around four principles: one runnable entry point for every maintained algorithm family, one normalized data-resolution layer for public and private assets, one configuration and output format for experiments, and one native Python codebase with tests rather than wrappers around obsolete scripts.

The maintained benchmark includes four algorithm families. The classical baseline ports a legacy handcrafted descriptor pipeline into Python using patch-wise local binary patterns and histogram-of-oriented-gradients features. The KinVer-style path consumes bundled feature matrices, applies Fisher-style feature filtering, PCA, an MNRML-like projection, and SVM fusion. The native GAE family preserves standard and multiview relation-learning modules over paired feature matrices stored in MAT-file format but loaded entirely through Python. The family-deep path is a native Siamese-style model with shared transforms, logging, and evaluation code for KinFaceW and FIW layouts.

Data normalization is a major engineering contribution of the migration. KinFaceW is relatively straightforward to resolve, but the local FIW FIDs export required more defensive logic. The new loader resolves pair metadata against the repo-local hierarchy, repairs certain metadata-to-image mismatches within a family branch when possible, and deterministically drops rows whose local crop is absent. This turns messy real data into documented protocol behavior rather than silent failure.

The toolkit also includes a private-dataset adapter for the in-house collection. The adapter inventories subsets, infers family-group structure where possible, and exports pair-manifest CSV files without placing the dataset under version control. This makes it possible to study private family archives within the same workflow while keeping the public repository clean and legally safe.

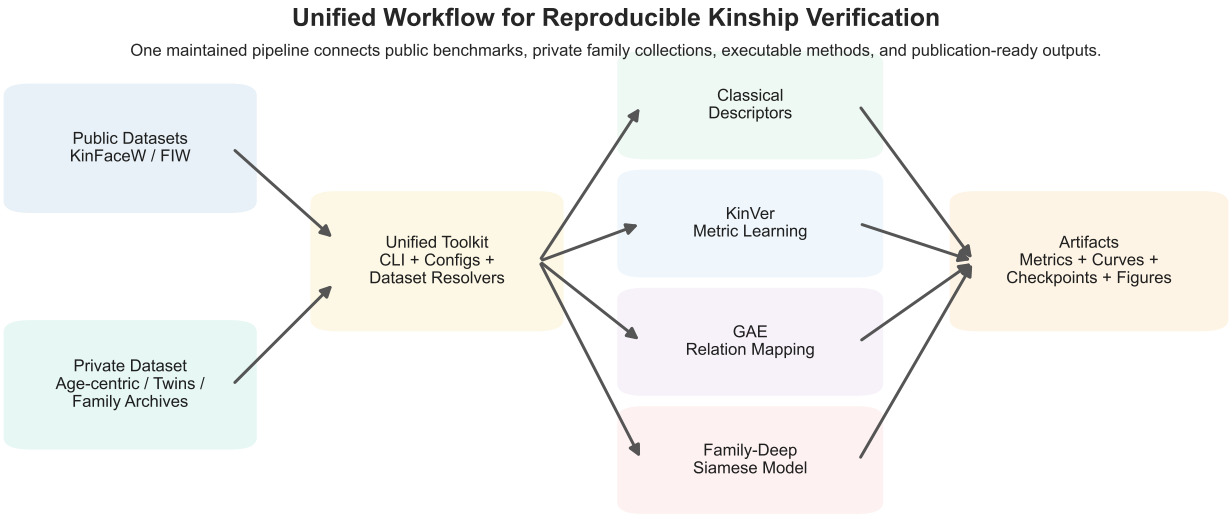

Fig. 2. Unified toolkit workflow from dataset resolution to reproducible experimental artifacts.

TABLE II
COMPOSITION OF THE IN-HOUSE DATASET.

Category	Fam.	People	Images	Focus
Age-centric kinship	65	203	962	Age-gap and age-progression analysis
Identical twins	102	292	434	Hard-negative and resemblance studies
Named family archives	4	48	322	Rich family context
Custom family collections	49	262	600	Broader local diversity
Total	220	805	2,318	Non-public research asset

IV. EXPERIMENTAL PROTOCOL

To keep the paper grounded in fully verified runs, we report only experiments executed through the maintained Python toolkit. Four public-facing experiment groups were used in this paper: (1) KinFaceW-II father-son verification using the bundled five-fold metadata (500 pairs) for the direct comparison across classical, KinVer, and family-deep; (2) a local FIW FIDs father-son export using repo-local train and validation CSVs, which yielded 66,346 training pairs and 24,072 validation pairs after image-resolution checks; (3) a native GAE validation run on a paired KinFaceW feature matrix stored in MAT-file format, containing 3,840 paired samples with 768 features per side; and (4) private-dataset staging runs over the local in-house collection, which currently covers 10 subsets and 2,318 images.

For the FIW experiment, 2,146 training pairs and 602 validation pairs were skipped because the local export did not contain a resolvable crop for at least one side of the pair. The classical path used the KinFaceW-II father-son five-fold protocol with k-fold evaluation. The KinVer path used bundled VGGFace and VGG-F features, Fisher-style filtering, PCA, an MNRML-style projection, and SVM fusion over five folds with four iterations and neighborhood size six. The family-deep path used a compact Siamese network under CPU-only execution. For KinFaceW-II the model was evaluated over five folds after a one-epoch schedule; for FIW it was trained for one epoch on the father-son relation and evaluated on the corresponding validation split. The GAE runs used ten epochs with `numfac = 16` and `nummap = 8`.

Fig. 3. Representative KinFaceW-II pairs illustrating appearance variation in kinship verification.

V. Results

A. KinFaceW-II Father-Son Verification

The reproduced KinVer path is the strongest performer by a large margin. This is consistent with the historical strength of task-specific metric learning when paired with robust face backbones and score fusion. As shown in Fig. 4, the KinVer advantage is not driven by a single favorable split; it remains strong across all five folds. Under our verified settings, the classical handcrafted baseline is essentially at chance, which confirms that the baseline is reproducible but not competitive on this setting. The deep Siamese baseline also remains near chance under the current one-epoch CPU-only reproduction. Rather than arguing against deep kinship verification in general, this result highlights a reproducibility lesson: making a deep pipeline runnable is not the same as fully optimizing it.

B. FIW Father-Son Validation on the Local FIDs Export

Although the thresholded classification result is weak, the AUC is meaningfully above chance, indicating that the model learned nontrivial ranking information even though its default threshold produced poor binary decisions. Fig. 6 makes this contrast explicit by showing that FIW is much stronger in ranking than in thresholded verification under the present one-epoch setting. From a benchmark perspective, the larger value of this result is that the maintained FIW loader, deep runner, and evaluator now operate end to end on a substantially larger and messier real-world family dataset than KinFaceW-II.

C. Native Gated Autoencoder Validation

The standard GAE achieved slightly lower reconstruction error than the multiview version on this KinFaceW matrix slice. Since no downstream verifier was trained on the mapped features within the present paper, these values should be interpreted as implementation-validation results rather than final recognition claims. Their importance is that the gated relation-learning family is now natively runnable in the same environment as the verification pipelines.

D. Private Dataset Staging

The private-dataset adapter was used to inventory the in-house collection and export pair manifests for future work. On the current workstation copy, the toolkit detected 10 subsets, 220 inferred family groups, 805 people, and 2,318 images. Within a representative identical-twin subset, the exported manifest contained 2,373 candidate pairs, including 2,044 positive pairs and 329 negative pairs. These numbers do not yet form a public benchmark, but they show that the toolkit can convert a locally curated family archive into machine-usable research assets without exposing private data in the public repository.

TABLE III
SUMMARY OF VERIFIED RESULTS FROM THE MAINTAINED TOOLKIT.

Dataset / Task	Method	Setting	Reported Metrics	Main Result
KinFaceW-II father-son	Classical patch descriptor + SVM	5-fold k-fold protocol	Mean accuracy	50.2%
KinFaceW-II father-son	KinVer (PCA + MNRML + SVM fusion)	5 folds, 4 iterations, $k = 6$	Mean accuracy	85.0%
KinFaceW-II father-son	Family-deep compact Siamese model	1 epoch, 5 folds	Mean accuracy, recall, precision, F1, AUC	49.6% mean acc.
FIW father-son local FIDs export	Family-deep compact Siamese model	1 epoch, validation split	Accuracy, recall, precision, F1, AUC	49.75% acc., 61.72% AUC
KinFaceW GAE matrix validation	Standard GAE	`numfac=16`, `nummap=8`, 10 epochs	Reconstruction error	0.6666
KinFaceW GAE matrix validation	Multiview GAE	`numfac=16`, `nummap=8`, 10 epochs	Reconstruction error	0.6833
Private dataset staging	Local pair-manifest export	Representative identical-twin subset	Pair inventory	2,373 pairs (2,044 positive / 329 negative)

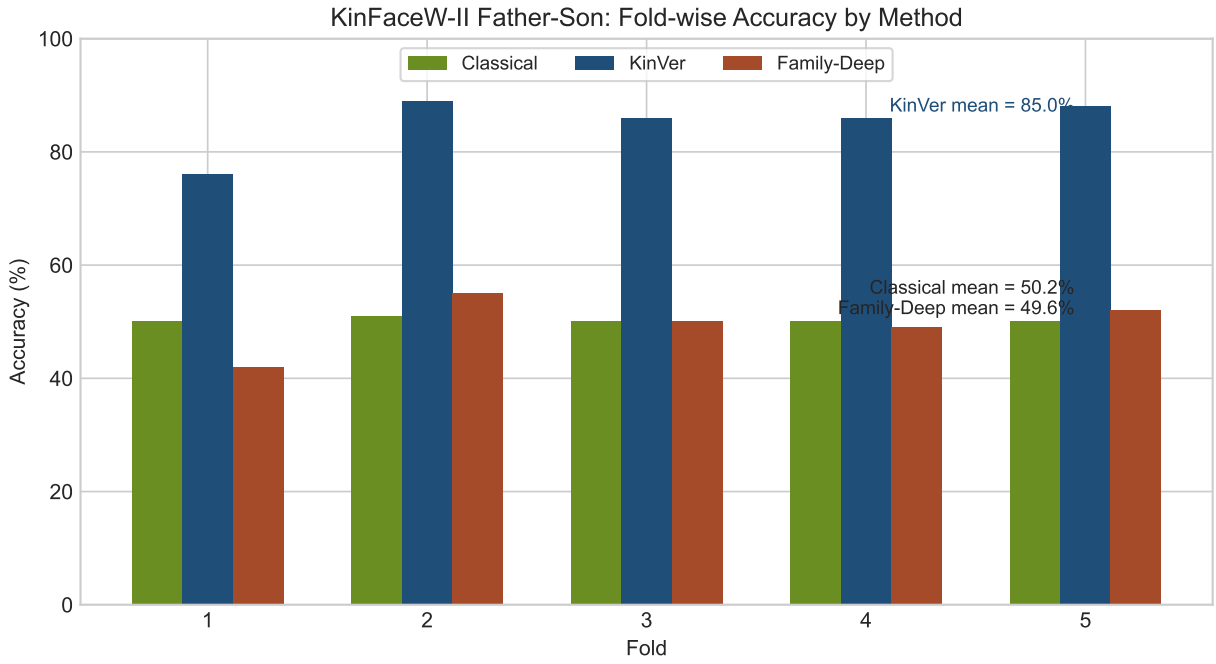

Fig. 4. Fold-wise KinFaceW-II father-son accuracy for the three directly comparable verification families.

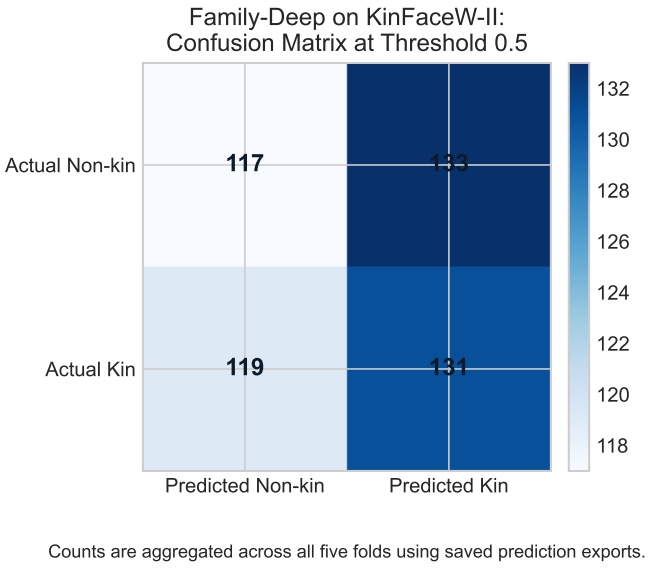

Fig. 5. Family-deep confusion matrix on KinFaceW-II father-son verification at threshold 0.5.

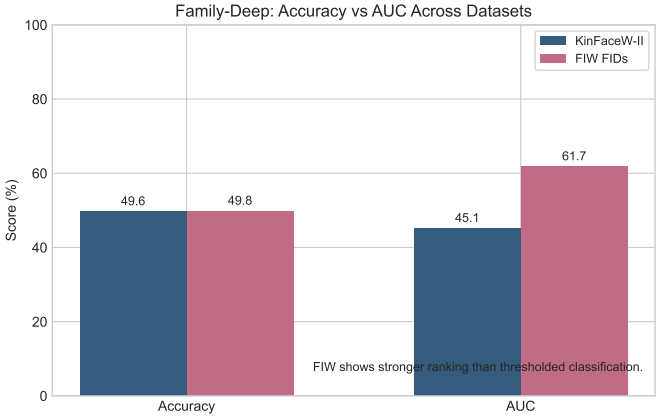

Fig. 6. Family-deep accuracy and AUC on KinFaceW-II and the local FIW export.

VI. Discussion

Three findings are worth emphasizing. First, executable comparison is now possible across historically separate kinship paradigms. Before migration, the algorithms existed as disconnected artifacts with incompatible dependencies, paths, and output formats. After migration, they can be run through the same interface and inspected through the same reporting layer.

Second, the strongest verified public result in the current benchmark comes from the KinVer-style metric-learning family rather than from the newly ported deep baseline. This is a useful reminder that older kinship-specific metric-learning approaches remain highly competitive when strong facial representations are available.

Third, data normalization is as important as algorithm migration. The FIW export used here was not immediately runnable under naive assumptions. The new loader's ability to resolve some metadata inconsistencies and document unresolved rows is itself a meaningful contribution because it turns messy local data into repeatable protocol behavior.

The in-house dataset further broadens the value of the project. Public datasets such as KinFaceW and FIW are indispensable, but they constrain research to the relations and capture conditions already present in those benchmarks. The local collection adds age-centric diversity, named-family structure, and a rare identical-twins subset. Because the dataset was assembled from scratch and already integrates cleanly into the toolkit, it represents a meaningful non-public research asset even though it is not yet part of a finalized public evaluation protocol.

This study also has clear limitations. The direct public comparison is restricted to the father-son relation because that is the cleanest overlap among the fully validated executable paths. The deep baseline was intentionally trained for only one epoch on CPU, so its results are reproducibility baselines rather than tuned ceilings. The GAE modules were validated through reconstruction-oriented runs rather than through a full downstream verifier. Finally, the private dataset is currently integrated as a local research asset and staging source, not as a released benchmark.

VII. Conclusion

The central result of this work is not a single new verifier, but a usable benchmark infrastructure for the field. By bringing classical handcrafted baselines, KinVer-style metric learning, native gated autoencoders, and a deep Siamese path into one maintained Python toolkit, we replace a fragmented software landscape with an executable research platform. The verified experiments show that KinVer remains a strong public baseline on KinFaceW-II, that the current deep path is stable but undertrained, and that the FIW FIDs pipeline now runs end to end under realistic local data conditions. Just as importantly, the toolkit can stage a substantially richer non-public family collection containing age-centric and identical-twin material, extending the project's experimental reach beyond standard public benchmarks. Together, these outcomes provide a credible foundation for stronger future studies, including better-calibrated deep models, broader kin relations, and eventual protocol formalization over the in-house dataset.

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